

Insights into the beneficial effect of caloric/ dietary restriction for a healthy and prolonged life

Over the last several years, new evidence has kept pouring in about the remarkable effect of caloric restriction (CR) on the conspicuous bedfellows- aging and cancer. Through the use of various animal models, it is now well established that by reducing calorie intake one can not only increase life span but, also, lower the risk of various age related diseases such as cancer. Cancer cells are believed to be more dependent on glycolysis for their energy requirements than normal cells and, therefore, can be easily targeted by alteration in the energy-metabolic pathways, a hallmark of CR. Apart from inhibiting the growth of transplantable tumors, CR has been also shown to inhibit the development of spontaneous, radiation, and chemically induced tumors. The question regarding the potentiality of the anti-tumor effect of CR in humans has been in part answered by the resistance of a cohort of women, who had suffered from anorexia in their early life, to breast cancer. However, human research on the beneficial effect of CR is still at an early stage and needs further validation. Though the complete mechanism of the anti-tumor effect of CR is far from clear, the plausible involvement of nutrient sensing pathways or IGF-1 pathways proposed for its anti-aging action cannot be overruled. In fact, cancer cell lines, mutant for proteins involved in IGF-1 pathways, failed to respond to CR. In addition, CR decreases the levels of many growth factors, anabolic hormones, inflammatory cytokines, and oxidative markers that are deregulated in several cancers. In this review, we discuss the anti-tumor effect of CR, describing experiments done in vitro in tumor models and in vivo in mouse models in which the tumor was induced by means of radiation or chemical exposure, expressing oncogenes or deleting tumor suppression genes. We also discuss the proposed mechanisms of CR anti-tumor action. Lastly, we argue the necessity of gene expression studies in cancerous versus normal cells upon CR.